

Abstract No.

Project Proposal: Rare Fusions in Pediatric Low Grade Glioma – a LOGGIC and LOGGIC-Core-BioClinical Data Bank Analysis

Isabella Horst^{1,}, Philipp Sievers^{2,3}, Emily C. Hardin^{4,5}, Julia Großer¹, Svea Horn¹, Isabel Fernandes Arroiteia¹, Thomas Brückner¹, Felix Sahm^{2,4}, Brigitte Bison⁶, Till Milde^{4,5}, Cornelis van Tilburg^{4,5}, David Capper⁷, Simone Schmid⁸, Arend Koch⁹, Olaf Witt^{4,5}, David T W Jones^{10,11}, Pablo Hernáiz Driever¹, and Theresa M Thole-Kliesch¹*

¹Charité-Universitätsmedizin Berlin, German HIT-LOGGIC-Registry for Low Grade Glioma in Children and Adolescents, Berlin, Germany

²Department of Neuropathology, Institute of Pathology, University Hospital Heidelberg, 69120, Heidelberg, Germany

³Clinical Cooperation Unit Neuropathology, German Cancer Consortium, German Cancer Research Center, 69120, Heidelberg, Germany

⁴Hopp Children's Cancer Center Heidelberg (KiTZ), Heidelberg, Germany

⁵Clinical Cooperation Unit Pediatric Oncology, German Cancer Research Center (DKFZ), German Cancer Consortium (DKTK), Heidelberg, Germany

⁶Department of Neuroradiology, University Hospital of Augsburg, Augsburg, Germany

⁷German Cancer Consortium (DKTK), Partner Site Berlin, German Cancer Research Center (DKFZ), Heidelberg, Germany

⁸Department of Neuropathology, Charité - Universitätsmedizin Berlin, Berlin, Germany

⁹12 Department of Neuropathology, Charité - Universitätsmedizin Berlin, Berlin, Germany

¹⁰National Center for Tumor Diseases (NCT), NCT Heidelberg, a Partnership Between DKFZ and Heidelberg University Hospital, Heidelberg, Germany

¹¹German Cancer Research Center (DKFZ), Im Neuenheimer Feld 280, 69120, Heidelberg, Germany

*Presenting author

Molecular pathology in routine diagnostic workup of pediatric low grade glioma (pLGG) tissue currently redefines the disease. Genetic workup of pLGG tumor tissue within the LOGGIC and LOGGIC Core registries includes an epigenetic profiling, NGS sequencing of 201 brain tumor-relevant genes and RNA sequencing. This offers to detect driver alterations in 97% of pLGG and therefor sets the stage for personalized medicine for the majority of pLGG patients. KIAA1549::BRAF fusions compose 50-60% of detected alterations, while rare fusions including FGFR1 fusions compose with 10-20% the second largest group of genetic alterations.

Rare fusions beyond KIAA1549::BRAF to date have not yet been systematically described in pLGG glioma. Our aim is to offer clinicians of patients with rare fusions a reference on what to expect regarding clinical behavior of the tumor and treatment efficacies.

Of 2700 patients registered in the HIT-LOGGIC registry between April 2019 and March 2025, 1028 were also registered in the LOGGIC Core bioclinical databank. 548 tumors received mRNA sequencing. Via mRNA sequencing 97 patients with tumors bearing non-KIAA1549::BRAF, non-FGFR fusions were identified as rare-fusion cohort. The most common fusion partners involved were BRAF (29), MYB(L) (16), RAF1 (13), NTRK 1-3 fusions (15), MYB::QKI (8), and GOPC::ROS (5). Rare gene fusions with specific fusion partners correlated with defined histologic entities of pLGG. Rare BRAF fusions mainly occurred in pilocytic astrocytoma, while MYB::QKI fusions



Supported by SciSerTec

Web: <http://www.scisertec.de>

occurred in angiocentric glioma or MYB/MYBL1-altered diffuse astrocytoma. BRAF breakpoints of rare BRAF fusions were mostly equivalent to canonical BRAF breakpoints described for KIAA1549::BRAF. However the frequency distribution differed significantly, with rare breakpoints in the context of KIAA1549::BRAF being the most common breakpoints among rare BRAF fusions.

Here, we systematically describe rare fusions in pLGG identified via mRNA sequencing and link molecular pathology to clinical features and patient courses, including sensitivity to different therapeutic regimen.